ARE IMPROPER PRIORS OBJECTIVE IN STATISTICAL INFERENCE?

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1. Introduction

In a general statistical problem, the goal is to make inferences about an unknown parameter θ. Depending on the problem, one may have particular beliefs on the parameter, in which case a *prior distribution* may be imposed on the parameter space Θ. In situations where such beliefs are weak or unavailable, it is a "common practice" to use *default priors* to mitigate personal beliefs and subjectivity that interfere with statistical inferences.

In the realm of default priors, a commonly used class of default priors is the class of improper priors. While the formal definition is given in Section 3, the defining feature of improper priors is that they do not integrate to one over Θ. That is, they do not properly define a valid probability measure on Θ, the parameter space. A canonical example is the uniform prior on Θ, which is constructed as $\pi(\theta) \propto 1$ for every value in the parameter space. Other popular choices include Jeffreys and reference priors, which will form a central focus in our paper. These are common choices for statisticians when choosing an appropriate prior for their statistical problem, mainly due to their simplicity and ability to produce, so-called, "objective" inferences through properties such as invariance.

However, this raises the fundamental question: if π is an improper prior on Θ, then it does not define a valid probability distribution on the parameter space Θ. In other words, can it still be meaningfully interpreted as a valid probability-based measure of belief about θ?

In this paper, we analyze this issue through classical paradoxes, Evans' framework of relative belief ratios, and prior-data conflicts. We will also discuss the issue where improper prior choices are generally not falsifiable via such methods. This, in the spirit of Karl Popper's framework of falsifiability, also raises some concerns on whether or not statistical inferences based on unfalsifiable components such as improper priors are valid scientific practice.

2. Acknowledgments

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3. Notations and definitions

In this section, we establish the notation and definitions that will be used throughout the paper. Throughout, let $\Theta, \mathcal{X}$ denote the parameter space and the sample space respectively.

Definition 3.1 (Proper and Improper Priors)**.** Let $\Theta \subseteq \mathbb{R}^d$. Then, a function $\pi : \Theta \to [0, \infty)$ is called a **proper** prior on Θ if

$$\int_\Theta \pi(\theta) d\theta = 1$$

It is called **improper** if it satisfies

$$\int_\Theta \pi(\theta) d\theta = \infty \text{ or undefined}$$

Definition 3.2 (Posterior Distributions)**.** Let $\Theta \subseteq \mathbb{R}^d$. Let $\pi(\theta)$ be a prior on Θ. Suppose that data x is observed with a likelihood function $f(x|\theta)$. Then, the **posterior** distribution of θ is defined as

$$\pi(\theta|x) = \frac{\pi(\theta) f(x|\theta)}{\int_\Theta \pi(\theta) f(x|\theta) d\theta}$$

provided that $0 < \int_\Theta \pi(\theta) f(x|\theta) d\theta < \infty$.

Definition 3.3 (Prior Predictive Densities)**.** Following the setup from Definition 3.2, we will call

$$m(x) = \int_\Theta \pi(\theta) f(x|\theta) d\theta$$

the **prior predictive density** of x.

Notation 3.4 (Sample mean)**.** We will let $\overline{X_n}, \overline{x_n}$ denote the random sample mean and the realized sample mean respectively of a sample of size n.

4. Classical Improper Priors

In this section, we will explore some common choices of improper priors used in practice, as well as why they are often considered an "objective" choice in inference.

4.1. Uniform Priors.

In many applied statistical contexts, when one has no particular beliefs and information on Θ, a common choice of improper prior would be the uniform prior.

Definition 4.1 (Uniform prior)**.** Let $\Theta \subseteq \mathbb{R}^d$ be the parameter space. Then, $\pi : \Theta \to [0, \infty)$ is called a **uniform** prior on Θ if

$$\pi(\theta) \propto 1, \forall \theta \in \Theta$$

While this seems like a reasonable choice since this represents that one has no preference among values of θ in Θ, this choice fails because of one fundamental issue: inconsistency under reparameterization. Here is a simple example.

Example 4.2 (Uniform Prior Failure in Reparameterization)**.** Let $\Theta = (0, \infty)$ be the parameter space. Suppose a uniform prior is imposed on Θ. That is,

$$\pi(\theta) = 1, \forall \theta \in \Theta$$

Let $\phi : \Theta \to \mathbb{R}$ be a bijection where $\Theta = (0, \infty)$, defined as

$$\phi(\theta) = \log(\theta)$$

Then, by change of variables, one would obtain the prior of ϕ as

$$\pi(\phi) = \pi(\theta)\left|\frac{d\theta}{d\phi}\right| = \pi(\theta)\left|\frac{d}{d\phi}e^{\phi}\right| = 1 \cdot e^{\phi} = e^{\phi}$$

Thus, $\pi(\phi)$ increases as $\phi \to \infty$. More importantly, it is no longer a flat prior. $\square$

Even while ignoring the issue of whether or not a uniform improper prior represents beliefs, Example 4.2 demonstrates that under simple, bijective reparameterizations of parameter values, the induced prior changes under reparameterization. For a prior to be considered "objective", it should be invariant under relabeling of parameters, namely, bijective transformations on Θ.

To resolve this issue of invariance [1] under reparameterization, a rather popular choice in practice is to use the Jeffreys prior, which was originally proposed by Harold Jeffreys [Jef86].

4.2. **Jeffreys Priors.**

Definition 4.3 (Jeffreys Priors)**.** Let $\Theta \subseteq \mathbb{R}^d$ be the parameter space. A prior $\pi : \Theta \to [0, \infty)$ is called the **Jeffreys** prior on Θ if

$$\pi(\theta) \propto \sqrt{\det(I(\theta))}, \forall \theta \in \Theta$$

where $I(\theta)$ denotes the Fisher information matrix for a single observation, defined as $I(\theta) = \mathbb{E}_\theta\left[(\nabla \log f_\theta(x))(\nabla \log f_\theta(x))^T\right]$, provided that $I(\theta)$ exists and is finite.

We now show its property of being invariant under bijective reparameterizations in a simple case, that is, in $\mathbb{R}$.

Theorem 4.4 (Jeffreys Prior Invariance under Reparameterizations)**.** *Let $\Theta \subseteq \mathbb{R}$ be a parameter space. Let $\{f_\theta(x) : \theta \in \Theta\}$ be the model. Let $\pi(\theta) \propto \sqrt{I(\theta)}$. Let $\psi : \Theta \to \Phi$ be a $\mathcal{C}^1$ bijective mapping, where $\Phi := \psi(\Theta) = \{\psi(\theta) : \theta \in \Theta\}$, and define $\phi = \psi(\theta)$. Then,*

$$\pi(\phi) \propto \sqrt{I(\phi)}$$

Proof. By definition,

$$I(\phi) = \mathbb{E}_\phi\left[\left(\frac{d}{d\phi}\log f_\phi(x)\right)^2\right]$$

where $f_\phi(x) = f_{\psi^{-1}(\phi)}(x)$. Let $\theta = \psi^{-1}(\phi)$. Then, by defining a pull-back on Φ via ψ and by chain rule, we get

$$\frac{d}{d\phi}\log f_\phi(x) = \frac{d}{d\phi}\log f_{\psi^{-1}(\phi)}(x) = \frac{1}{f_\theta(x)}\frac{d}{d\theta}f_\theta(x)\frac{d\psi^{-1}(\phi)}{d\phi} = \frac{1}{f_\theta(x)}\frac{d}{d\theta}f_\theta(x)\frac{d\theta}{d\phi}$$

Since $f_\phi(x) = f_{\psi^{-1}(\phi)}(x)$, expectations coincide under θ and ϕ, so it follows that

$$I(\phi) = \mathbb{E}_\phi\left[\left(\frac{1}{f_\theta(x)}\frac{d}{d\theta}f_\theta(x)\frac{d\theta}{d\phi}\right)^2\right] = \mathbb{E}_\phi\left[\left(\frac{d}{d\theta}f_\theta(x)\frac{d\theta}{d\phi}\right)^2\right] = \left(\frac{d\theta}{d\phi}\right)^2\mathbb{E}_\theta\left[\left(\frac{d}{d\theta}f_\theta(x)\right)^2\right] = \left(\frac{d\theta}{d\phi}\right)^2 I(\theta)$$

[1] Whether or not being unable to be invariance under reparameterization is really an issue, is open to interpretation.

Thus, it follows immediately that

$$\sqrt{I(\phi)} = \sqrt{\left(\frac{d\theta}{d\phi}\right)^2 I(\theta)} = \left|\frac{d\theta}{d\phi}\right| \sqrt{I(\theta)}$$

By change of variables,

$$\pi(\phi) = \pi(\theta) \left|\frac{d\theta}{d\phi}\right|$$

Since $\pi(\theta) \propto \sqrt{I(\theta)}$,

$$\pi(\phi) = \pi(\theta) \left|\frac{d\theta}{d\phi}\right| \propto \sqrt{I(\theta)} \left|\frac{d\theta}{d\phi}\right| = \sqrt{I(\phi)}$$

which completes the proof. □

In fact, in full generality, Jeffreys prior is invariant under bijective, differentiable reparameterizations in $\mathbb{R}^d$, which resolves the issue identified in Example 4.2. Moreover, structurally, the inclusion of Fisher information in Jeffreys prior introduces considerations of the geometry of the parameter space. That is, notice that the value of Jeffreys priors increases with $I(\theta)$, meaning that it places greater emphasis on values of θ in Θ that are comparatively more identifiable.

One notable property of Jeffreys prior is that sometimes, the results it produces match frequentist results.

Example 4.5 (Jeffreys Prior on Location Normals)**.** Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$, where $\theta \in \mathbb{R}$ is unknown. Then, the log-likelihood for θ would be

$$\ell(\theta; x_1, \dots, x_n) = \frac{-n}{2}(\log(2\pi)) - \frac{\sum_{i=1}^n (x_i - \theta)^2}{2}$$

Hence,

$$I(\theta) = \mathbb{E}_\theta\left[\left(\frac{d}{d\theta}\ell(\theta; x_1, \dots, x_n)\right)^2\right] = \mathbb{E}_\theta\left[\left(\sum_{i=1}^n (X_i - \theta)\right)^2\right] = \sum_{i=1}^n \mathrm{Var}_\theta\left[X_i\right] = n$$

since $\mathrm{Var}[X_i] = 1, \forall i \in \{1, \dots, n\}$. Thus,

$$\pi_{\text{Jeff}}(\theta) \propto \sqrt{n} \propto 1$$

While this Jeffreys prior is remarkably simple, the results it produces, in this specific case, matches frequentist results in some particular aspects. For instance, in the normal model in this context, the realized MLE is given as $\hat{\theta}_{\text{MLE}} = \overline{x_n}$, the sample mean. Notice that under the Jeffreys prior on θ, the posterior distribution, obtained via Definition 3.2 and by completing the square in θ, is given as

$$\pi(\theta | x_1, \dots, x_n) = \mathcal{N}\left(\overline{x_n}, \frac{1}{n}\right)$$

And it follows that

$$\mathbb{E}_{\pi(\cdot|x_1,\dots,x_n)}[\theta] = \overline{x_n} = \hat{\theta}_{\text{MLE}}$$

Hence, the realized MLE coincides with the posterior mean. Moreover, since $\overline{X_n} \sim \mathcal{N}\left(\theta, \frac{1}{n}\right)$, the realized frequentist size $1-\alpha$ confidence interval for θ is

$$\left(\overline{x_n} - \frac{z_{\alpha/2}}{\sqrt{n}}, \overline{x_n} + \frac{z_{\alpha/2}}{\sqrt{n}}\right)$$

Since the posterior distribution for θ is $\mathcal{N}\left(\overline{x_n}, \frac{1}{n}\right)$, it turns out also that the $1-\alpha$ credible interval for θ is also

$$\left(\overline{x_n} - \frac{z_{\alpha/2}}{\sqrt{n}}, \overline{x_n} + \frac{z_{\alpha/2}}{\sqrt{n}}\right)$$

where $z_{\alpha/2}$ is the $\alpha/2$ critical value of $\mathcal{N}(0,1)$. The results above match frequentist results exactly. As a side note, in this example, the prior on θ also coincides with the right Haar prior on $(\mathbb{R},+)$. □

While this brings us closer to constructing an "objective" improper prior in Bayesian contexts, as Berger and Bernardo noted in their paper in 1991 on the developments of reference priors [BB91], Jeffreys priors sometimes require "ad-hoc" modifications in higher dimensional settings. For instance, in a statistical problem where there are nuisance parameters, the prior of the parameter of interest will differ depending on the method of construction that we use. Here is an example to illustrate this.

Example 4.6 (Marginalizations of Jeffreys Priors in 2D)**.** Suppose $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}, \sigma > 0$, both unknown. Suppose our parameter of interest is μ, and σ is a nuisance parameter.

Treating σ as fixed: Then, notice that

$$f(x_1, \ldots, x_n; \mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\sum_{i=1}^n (x_i-\mu)^2}{2\sigma^2}\right)$$

And hence the log-likelihood in μ will be

$$\ell(\mu; x_1, \ldots, x_n, \sigma) = \frac{-n}{2}\left(\log(2\pi\sigma^2)\right) - \frac{\sum_{i=1}^n (x_i-\mu)^2}{2\sigma^2}$$

$$I(\mu) = \mathbb{E}_\theta\left[\left(\frac{d}{d\mu}\ell(\mu; x_1, \ldots, x_n, \sigma)\right)^2\right] = \mathbb{E}_\theta\left[\left(\frac{-1}{2\sigma^2}\sum_{i=1}^n 2(X_i-\mu)\right)^2\right] = \frac{1}{\sigma^4}\mathbb{E}_\theta\left[\left(\sum_{i=1}^n (X_i-\mu)\right)^2\right]$$
$$= \frac{1}{\sigma^4}\left[\mathrm{Var}_\theta\left(\sum_{i=1}^n (X_i-\mu)\right)\right] = \frac{1}{\sigma^4}\sum_{i=1}^n \mathrm{Var}_\theta\left(X_i\right) = \frac{n}{\sigma^2}$$

by independence. Thus, the Jeffreys prior in this case would be

$$\pi_{\text{direct}}(\mu) \propto \sqrt{\frac{n}{\sigma^2}} = \frac{\sqrt{n}}{\sigma} \propto \frac{1}{\sigma}$$

Marginalizing over the joint Jeffreys prior: Before we could marginalize, we first consider the joint Jeffreys prior on (μ, σ). Since the data is normally distributed, by direct computation, the Fisher information matrix is given by

$$I(\mu,\sigma) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{2n}{\sigma^2} \end{pmatrix}$$

Thus,

$$\pi_{\text{Jeff}}(\mu,\sigma) \propto \sqrt{\det I(\mu,\sigma)} = \sqrt{\frac{2n^2}{\sigma^4}} \propto \frac{1}{\sigma^2}$$

In particular, it is an improper joint Jeffreys prior on (μ,σ), and "marginalizing" this joint Jeffreys prior is not a well-defined operation. If one attempts to integrate out σ from the joint prior, we have

$$\int_0^\infty \pi_{\text{Jeff}}(\mu,\sigma)d\sigma = \int_0^\infty \frac{1}{\sigma^2}d\sigma = \infty$$

which shows that no meaningful prior on μ is obtained via "marginalizing" the joint Jeffreys prior in this specific construction. □

The construction in Example 4.6 exposes a fundamental inconsistency of the Jeffreys prior. Here, we have constructed two Jeffreys priors for μ, both of which are formally, valid constructions. However, we have showed that directly computing the Jeffreys prior on μ given that σ is fixed yields a closed form solution, yet attempting to marginalize the joint Jeffreys prior on (μ,σ) is not a well-defined operation since the joint Jeffreys prior is in itself improper. This violates yet another consistency principle that a "objective" improper prior on the parameter space should at least follow. That is, "objective" priors should, at minimum, yield consistent results independent of the method of construction.

It is to be noted that this phenomenon not only happens with Jeffreys priors, but also with general improper priors too. Such an example is used simply to illuminate the issue where priors depend on the "route" by which they are constructed.

This motivates the broader class of improper priors introduced by Bernardo and Berger, known as *reference priors*. Given that regularity conditions are satisfied, the Jeffreys prior is a special case of such priors as well.

4.3. **Reference Priors.** The construction of a reference prior is derived information-theoretically, thus, we begin by introducing some definitions.

Definition 4.7 (Kullback-Leibler/Logarithmic Divergence)**.** Let $p(\theta), q(\theta)$ be probability density functions on $\Theta \subseteq \mathbb{R}^d$. Then, the **KL-divergence/logarithmic** divergence of $q(\theta)$ from $p(\theta)$ is defined as

$$D_{\text{KL}}(p\|q) = \int_\Theta p(\theta)\log\left(\frac{p(\theta)}{q(\theta)}\right)d\theta$$

when this integral is well-defined and finite.

While the KL-divergence is not a metric on the function space of probability densities, it still provides, informally speaking, a fair "measure of distances" between densities p and q on Θ. To Bernardo and Berger, their formulation of reference prior hinges on the principle where, roughly speaking, an optimal prior should maximize the expected KL-divergence between the posterior and the prior to let the data have maximal influence on the posterior. Thus, to some, this is an "objective" choice of prior since it lets them provide judgments using data when we are ignorant about information on θ. We will briefly introduce its formal construction below (See [BBS09b]).

Definition 4.8 (Expected Logarithmic convergence)**.** Let the model be $\{f_\theta(x) : \theta \in \Theta, x \in \mathcal{X}\}$ where $\Theta, \mathcal{X} \subseteq \mathbb{R}^d$. Let π be a prior on Θ. Then, let $\{\Theta_i\}_i$ be a sequence of compact subsets of Θ such that $\bigcup_i \Theta_i = \Theta$. Let π_i be the *truncated* prior on Θ_i. Then, we say that the sequence of posteriors on Θ_i, $\{\pi_i(\theta|x)\}_i$ is **expected logarithmic convergent** to $\pi(\theta|x)$ if

$$\lim_{i\to\infty} \int_{\mathcal{X}} D_{\mathrm{KL}}(\pi(\cdot|x)\|\pi_i(\cdot|x))p_i(x)dx = 0$$

and $p_i(x)$ is defined as

$$p_i(x) = \int_{\Theta_i} f_\theta(x)\pi_i(\theta)d\theta$$

Here, Definition 4.8 formalizes the notion of convergence between a sequence of posteriors restricted to compact subsets of Θ to our true posterior on Θ. In some sense, for a sequence of posteriors to converge logarithmically based on the definition above, they had to do so in expectation over $\mathcal{X}$ which is much stronger than, for example, pointwise convergence in practice, making it a suitable notion of convergence. In Bernardo and Berger's framework on reference priors, one of the criteria that it has to satisfy is that it has to be *permissible*, which incorporates this idea of convergence.

Definition 4.9 (Permissible priors)**.** Let π be a strictly positive prior density on Θ. Let $\{f_\theta(x) : \theta \in \Theta, x \in \mathcal{X}\}$ be the model. Then, we say that π is a **permissible prior** on Θ if:

(1) $\int_\Theta \pi(\theta)f_\theta(x)d\theta < \infty$
(2) There exist a sequence of priors $\{\pi_i\}_i$ on Θ_i, where Θ_i is a compact subset of Θ, such that $\{\pi_i(\theta|x)\}_i$ expected logarithmic convergence to $\pi(\theta|x)$.

While the goal of reference priors is to essentially "let the data speak the most", as noted by Berger, Bernardo and Sun's in their formalization of reference priors (See [BBS09b]), that in general, analysis on unbounded parameter spaces Θ are generally rather challenging. To address this, one instead considers a sequence of compact subsets of Θ and study the limiting behavior of the corresponding posteriors. Thus, by imposing the condition of permissibility on our priors, we ensure that this limiting behavior is met and well-defined.

Then, to formalize the idea of "letting the data speak the most", Berger, Bernardo, and Sun introduced the MMI (Maximizing Missing Information) property, which characterizes reference priors as those that maximize the expected information gain from data. The definition of MMI relies on the definition of *expected information*, which we define below.

Definition 4.10 (Expected Information)**.** Let $\mathcal{M} = \{f_\theta(x) : \theta \in \Theta, x \in \mathcal{X}\}$ be the model. Let π be a prior on Θ. Then, we define the **expected information** from one observation of the model, denoted as $I[\pi|\mathcal{M}]$, defined as

$$I[\pi|\mathcal{M}] = \int_{\mathcal{X}} D_{\mathrm{KL}}(\pi(\cdot|x)\|\pi(\cdot))p(x)dx$$

where

$$p(x) = \int_\Theta f_\theta(x)\pi(\theta)d\theta$$

Informally speaking, the expected information of an observation is a way for us to quantify, on average, the amount of information gained on θ after observing x. Intuitively, if we gained more information after x, then the prior and the posterior should "differ" more, which is precisely measured using KL-divergence. That is, the greater the average KL-divergence over $\mathcal{X}$ between the prior and the posterior, the larger the $I[\pi|\mathcal{M}]$. This provides the appropriate basis for defining the second criterion for reference priors, namely, the MMI property below.

Definition 4.11 (Maximizing Missing Information Property)**.** Let $\mathcal{M} = \{f_\theta(x) : \theta \in \Theta, x \in \mathcal{X}\}$ be a model, and let $\mathcal{P}$ be a class of prior densities on Θ such that

$$\int_\Theta f_\theta(x)\pi(\theta)\, d\theta < \infty.$$

Let $\mathcal{M}^k$ denote k independent repetitions of $\mathcal{M}$. Let $\{\Theta_i\}$ be an increasing sequence of compact subsets of Θ with $\bigcup_i \Theta_i = \Theta$. For every $\pi \in \mathcal{P}$, define π_i to be the *truncated* prior on Θ_i. Then, π is said to satisfy the MMI property if, for all $\pi' \in \mathcal{P}$, and for all Θ_i,

$$\lim_{k\to\infty} \left\{ I[\pi_i \mid \mathcal{M}^k] - I[\pi_i' \mid \mathcal{M}^k] \right\} \geq 0.$$

While this definition may appear technical, it conveys a simple idea. That is, a prior satisfies MMI when, among all of the possible priors with finite predictive densities, it gains the most information (information-theoretically) after observing data asymptotically under repeated sampling. This, in its core, is what Bernardo and Berger aimed to achieve: to choose a prior that maximizes the amount of learning purely by data. With the permissibility and MMI property being well-defined, here is the definition of reference priors.

Definition 4.12 (Reference priors (Bernardo, Berger, Sun))**.** Let π be a prior on Θ. Then, we say that $\pi(\theta)$ is a **reference prior** for the model $\mathcal{M}$ given the class of prior densities $\mathcal{P}$ if it satisfies both the MMI property and is permissible.

As hinted in the introduction of this subsection, Clarke and Barron ([CB94]) proved that under repeating sampling from the model and regularity conditions, the reference prior and the Jeffreys prior coincide. In addition, the construction of reference priors above possesses several desirable traits that may suggest that it is an "objective" choice over other improper priors, such as being invariant under sufficient statistics ([BBS09b]), and being invariant under reparameterization ([BBS09b]). In the case where the parameter space is finite, if we apply the theory of reference priors from above to construct it, as Bernardo and Smith proved in [BS94], it indeed yields a uniform prior on Θ.

However, as noted by Bernardo and Berger in [BBS09a], we usually make modifications in these contexts since such problems lack "structure". For instance, we would embed the problem in continuous settings, which in turn introduces "structure" based on their interpretation. However, an identical problem that we identified with reference priors arises, that is, if structural modifications are needed depending on the problem, this inherently would introduce some degree of subjectivity and challenges the "objectivity" of reference priors in a structural sense. More importantly, unlike other default priors from previous sections, it still remains notoriously hard to derive a reference prior for a given statistical problem even though an explicit procedure exists (See [BBS09b]).

This raises a fundamental question: even though such choice is information theoretically optimal, given its mathematical complexity, does this really meaningfully correspond to the notion of "objectivity" in statistical inference if its highly challenging to implement? In particular, if a prior cannot be effectively applied in practice, it is rather unclear whether it can still be assessed on its "objectivity" in an operational perspective. This still remains as an open problem concerning the trade-off between optimality and practicality.

5. Are these improper priors truly "objective"?

Before we analyze whether improper priors are even "objective", we briefly review the construction of reference priors, as given in Definition 4.12.

5.1. **Measure of "distance" in reference priors.** In Definition 4.12, notice that the MMI property relies on KL-divergence as a "measure of distance" to quantify the learning from data, which is reflected by the "difference" between the prior and the posterior. However, what justifies the choice of KL divergence as a "measure of distance between densities"?

Formally, KL-divergence, as mentioned previously, is not a metric on the function space $\mathcal{F} = \{p : \Theta \to [0, \infty) : \int_\Theta p(\theta)d\theta = 1\} \subset L_1(\Theta)$. Specifically, it fails the triangle inequality and the symmetry requirements that a metric should fulfill, which is to be expected since intuitively, KL-divergence measures expected losses (in log-scale) of our estimate with the true distribution asymmetrically. Thus, given the many choices of metrics in statistics, does the choice of the divergence in the definition introduce some degree of subjectivity? As an example, the total variation distance, d_{TV} between two probability densities $p(\theta), q(\theta)$ on Θ, defined as

$$d_{TV}(p\|q) = \frac{1}{2}\int_\Theta |p(\theta) - q(\theta)|d\theta$$

is a fully legitimate metric on $\mathcal{F}$ that prioritizes optimality in absolute distances (L_1-norm).

In the construction of reference priors, a possibility that KL-divergence is chosen as opposed to other metrics is likely justified due to KL-divergence being the canonical choice in statistical practice. As an example, standard statistical methods such as maximum likelihood estimation in inference and the EM (Expectation Maximization) algorithm in statistical computation all leverage KL-divergence in their constructions. Another reason that KL-divergence is a common choice in statistical inference arises when the model is misspecified. If $\{f_\theta(x) : \theta \in \Theta\}$ is our model for a given problem, a model is considered *misspecified* if the true data-generating distribution f_{True} is not in our model. It turns out that in this misspecified context, the MLE converges to the parameter $\theta' \in \Theta$ (also known as the pseudo-true parameter) such that $f_{\theta'}$ is the minimizer of the KL-divergence between f_θ for $\theta \in \Theta$ and f_{True} [2], which highlights the significance of KL-divergence in these contexts.

Nevertheless, in this sense, the specific choice of distance used to quantify the differences between densities in the construction of reference priors may not be fully considered "objective" since it implicitly depends on the choice of the notion of "optimality".

Even if this dependency is overlooked, reference priors expose a crucial flaw[3], namely, it depends on the ordering of parameters when constructing such priors.

5.2. **The ordering of parameters in reference priors.** As mentioned in a paper by Berger and Bernardo on ordered group reference priors on multinomial problems ([BB92]), in higher dimensional problems, one often follows an almost "algorithmic" approach when finding the reference prior on the parameter of interest. In general, the computational framework proposed by Bernardo and Berger was to reduce the problem to a sequential, one dimensional problem: First, compute the conditional reference prior conditioned on our

[2]Just to clarify due to the asymmetry of KL divergence, we are minimizing $D_{\text{KL}}(f_{\text{True}}\|f_\theta)$ for $\theta \in \Theta$.

[3]Similar to before, whether or not this is a flaw is open to interpretation.

parameter of interest, then find the marginal reference prior of our parameter of interest by integrating out all the nuisance parameters.

However, what if the parameter of interest is of very high dimensional? For instance, in the multinomial example exhibited in [BB92], they used a k-group approach where the components of the parameter of interest are separated into k disjoint groups, then, marginal reference priors are constructed sequentially on these k-groups. However, how does one determine the order in which we compute these marginal reference priors when there are multiple groups present? In [BB92], they ordered the parameters in their "order of interest". However, this choice is intrinsically subjective and more importantly, different orderings generally produce different results, since distinct priors normally result in distinct posteriors, leading to inferences that differ.

Therefore, this construction of reference priors as a potential remedy to the problem of constructing"objective informative priors" depends on arbitrary choices such as the ordering of parameters, which is not "objective" in the intrinsic definition of the term "objectivity". This, in turn, may lead to significant, varying results when parameters of very high dimensions are assessed, say, in setting like neural networks or high dimensional graphical models, where it is possible that there are numerous different orderings on the parameters.

5.3. **Jeffreys-Lindley Paradox.** Up to this point, the critiques of reference priors are primarily focused on the implementation and mathematical choices. We will now return to a simple setting introduced earlier, that is, Example 4.5, where we emphasized that a primary reason why Jeffreys prior is a prior that is widely considered is because of its ability to provide a potential reconciliation between Bayesian and frequentist approaches. However, is such reconciliation truly statistically plausible? We will investigate this issue through the canonical *Jeffreys Lindley paradox*. We first introduce several definitions.

Definition 5.1 (Relative Belief Ratios ([Eva16]))**.** Let $A, C \subseteq \Omega$ for some sample space Ω. Then, the *principle of relative belief* for an event A being true having observed event C is measured via the relative belief ratio

$$\mathrm{RB}(A|C) = \frac{\mathbb{P}(A|C)}{\mathbb{P}(A)}$$

where $\mathbb{P}(A) > 0$.

Definition 5.2 (Bayes Factors ([Eva16]))**.** Let $A, C \subseteq \Omega$ for some sample space Ω. Then, the Bayes factor $\mathrm{BF}(A|C)$ is defined as

$$\mathrm{BF}(A|C) = \frac{\mathrm{RB}(A|C)}{\mathrm{RB}(A^C|C)}$$

Example 5.3 (Jeffreys-Lindley Paradox)**.** We will be following the framework outlined in [Eva26]. Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$, and $\mu \in \mathbb{R}$ unknown. Then, we know that $\overline{X_n} \sim \mathcal{N}\left(\mu, \frac{1}{n}\right)$. Suppose we are interested in testing:

$$H_0 = \{0\}, H_a = \mathbb{R} \setminus \{0\}$$

Define $\mu_0 := 0$ conventionally. Let the test statistic $T(X)$ be defined as $T(X) = \sqrt{n}\overline{X_n}$ and suppose we have observed that $\sqrt{n}\overline{x_n} = 5$. Under a size $\alpha = 0.05$ test, this provides very strong evidence against H_0 under the frequentist framework since the two sided p-value would be $2 \cdot \Phi(-5)$ which is on the order of $10^{-7} \ll 0.05$. Fix $p \in (0,1)$ for the mixture weight. We will impose a spike-and-slab prior on μ defined by

$$\pi(\mu) = p\delta_{\mu_0} + (1-p)\pi_{\text{Jeff}}(\mu)$$

where δ_{μ_0} is the degenerate point mass on μ_0. By Example 4.6, we have shown that the prior on μ will be a uniform, flat prior on Θ. Let $\pi_{\text{Jeff}}(\mu) \propto 1$.

We will first compute $\mathbb{P}(H_0|\overline{x_n})$. By Bayes' theorem, it follows that

$$\mathbb{P}(H_0|\overline{x_n}) = \frac{f_{\mu_0}(\overline{x_n})\mathbb{P}(H_0)}{f(\overline{x_n})} = \frac{pf_{\mu_0}(\overline{x_n})}{f(\overline{x_n})}$$

We first compute $f_{\mu_0}(\overline{x_n})$. Since $\mu_0 = 0$, it follows that

$$f_{\mu_0}(\overline{x_n}) = \frac{1}{\sqrt{2\pi\frac{1}{n}}} \exp\left(-\frac{1}{2}\left(\frac{\overline{x_n}}{\frac{1}{\sqrt{n}}}\right)^2\right) = \frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}\left(\overline{x_n}\right)^2\right)$$

For the calculation of $f_\mu(\overline{x_n})$ under H_a, please refer to the appendix. By equation 6.4 in the appendix, we obtained that

$$f(\overline{x_n}) = 1$$

We now calculate $f(\overline{x_n})$. By the law of total probability, one obtains

$$f(\overline{x_n}) = f(\overline{x_n}|H_0)\mathbb{P}(H_0) + f(\overline{x_n}|H_a)\mathbb{P}(H_a) = p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right) + (1-p)$$

Hence,

$$\mathbb{P}(H_0|\overline{x_n}) = \frac{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right)}{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right) + (1-p)}$$

Similarly, it follows immediately that

$$\begin{aligned}\mathbb{P}(H_a|\overline{x_n}) = 1 - \mathbb{P}(H_0|\overline{x_n}) &= 1 - \frac{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right)}{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right) + (1-p)} \\ &= \frac{1-p}{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right) + (1-p)}\end{aligned}$$

Hence, the Bayes factor will be

$$\frac{\frac{\mathbb{P}(H_0|\overline{x_n})}{\mathbb{P}(H_a|\overline{x_n})}}{\frac{p}{1-p}} = \frac{p\left(\frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)\right)}{1-p} \cdot \frac{1-p}{p} = \frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}(\overline{x_n})^2\right)$$

Having observed $\sqrt{n}\overline{x_n} = 5$, this means that $n\overline{x_n}^2 = 25$ so we Bayes factor will be

$$\text{BF}(H_0|\overline{x_n}) = \frac{\sqrt{n}}{\sqrt{2\pi}} \exp(-12.5)$$

which goes to infinity as $n \to \infty$.

Thus, by the Jeffreys scale in [Eva26], we have decisive evidence in favour of H_0 in the Bayesian regime, which is a direct "disagreement" with the frequentist conclusion we obtained earlier. $\square$

As commented in Example 4.5, one of the principal justifications for why Jeffreys priors are considered "objective" is due to its ability to produce Bayesian results that agree with frequentist results in specific problems. Historically, frequentist approaches are viewed as "objective" due to the idea of repeated sampling and long run frequencies, and hence this trait is a major consideration when choosing an "objective", non-informative prior given a statistical problem by early statisticians. For instance, in 1963, Welch and Peers showed that up to order $\mathcal{O}(n^{-1})$, Jeffreys prior (or more generally *first order matching priors* as noted by Mukerjee and Reid ([MR99])), produces Bayesian credible intervals with correct frequentist covering probabilities ([WP63]). However, as we saw above, Jeffreys-Lindley paradox illustrates that such agreement between the Bayesian and frequentist results does not generally apply to hypothesis testing problems, indicating that an alignment of results between the Bayesian and frequentist approaches are regime-specific and are highly dependent on the nature of the problem (say, estimation or hypothesis testing, which are fundamentally different). In our view, Bayesian and frequentist methodologies significantly differ in their goals and hence such misalignment in results, and framework-based objectivity, should be rather unsurprising.

While the two paradigms fundamentally differ, as motivated by discussions with M. Evans, frequentist "checks" such as assessing for bias in favour via relative belief ratios, under the alternative hypothesis, is crucial when evaluating the quality of a given prior (See [EG21]). Mathematically, in our case of the Jeffreys-Lindley paradox, one could calculate

$$\sup_{\mu \in \{\mu' \in \mathbb{R} : d(\mu', \mu_0) \geq \delta\}} \mathbb{P}_\mu(\mathrm{RB}(\mu_0|X) \geq 1)$$

where $\delta > 0$ is fixed for identifiability between the null and the alternative, and $d(\cdot, \cdot)$ is assumed to be the Euclidean metric. Evans, in [Eva26], showed that in the case where an arbitrarily diffuse prior, say $\mathcal{N}(0, \tau_0^2)$ for some $\tau_0 > 0$ that is arbitrarily large, then under *any* alternative μ that is meaningfully distinguished from our null hypothesis, we will have bias in *favour* of the null hypothesis under repeated sampling. That is, the choice of an arbitrarily diffuse prior (which is asymptotically a uniform flat prior), produces evidence that supports the null hypothesis even though we know for sure, that the null hypothesis is false, which demonstrates the importance to enforce such validations on priors. However, this framework does not apply to improper priors, which to a greater extent challenges its reliability and validity of being an "objective" choice of priors in inference. Our discussion on its inability to undergo model checking and more importantly, to measure beliefs, continues in the subsequent sections.

5.4. **Measuring Beliefs.** In Bayesian inference, for a prior to be "objective", it should be used to accurately measure, or reflect, one's belief for an unknown parameter θ. For instance, if one places a $\mathcal{N}(0, 1)$ prior on θ, then, their beliefs on θ would be that θ is most likely around 0 and unlikely to be large. However, since improper priors do not integrate to 1, it is not a valid probability distribution for θ and hence is not an appropriate representation of belief.

This however, raises a further concern: If statistical evidence is measured via changes in belief using tools like relative belief ratios (Definition 5.1), then the use of improper priors as objective inference becomes questionable and uncertain. More specifically, since they are not proper tools to be used to measure belief probabilistically, subsequently, the definition of statistical evidence will be rather ambiguous and not well-defined in the case of improper priors. Above all, we will now show that in general, not only is it not a valid measure of belief, it also generally cannot be checked for compatibility with data.

5.5. **Prior-Data Conflict.** In a general statistical problem, we would generally like to have some diagnostics to check whether or not the prior or the model that we chose for the problem is compatible with what we observed. In current literature, Evans and Moshonov [EM06] proposed a method in which we could check for prior compatibility with data. That is, using the framework of *prior data conflict.* For a given model $\mathcal{M} = \{f_\theta(x) : x \in \mathcal{X}, \theta \in \Theta\}$ and a prior π on Θ, the method is as follows:

(1) Derive the prior-predictive distribution of X (See Definition 3.3). That is,
$$m(x) = \int_\Theta f(x|\theta)\pi(\theta)d\theta$$
(2) Given an observation x_0, compute the value of
$$M(m(X) \leq m(x_0))$$
where $M(\cdot)$ is the prior predictive measure induced by $m(x)$.
(3) If such value is small, then the observed value is statistically "rare" under our model hence we conclude with prior-data conflict, meaning our choice of prior may be inadequate in our problem.

Note that in practice, one would use the marginalized model with the minimal sufficient statistics rather than the full data. However, the defining feature of improper priors is that they do not integrate to 1 over Θ. That is, if π is improper, then adhering to Definition 3.1
$$\int_\Theta \pi(\theta)d\theta = \infty \text{ or undefined}$$
In a lot of cases, improper priors yield degenerate results when checking for prior data conflicts. Here is an example.

Example 5.4 (Prior Data Conflict: Flat Prior with Location Cauchy)**.** Let $\Theta = \mathbb{R}$ be the parameter space and impose a uniform prior $\pi(\theta) = 1$ on Θ. Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Cauchy}(\theta, 1)$ where $\theta \in \mathbb{R}$ unknown. Then,
$$m(x) = \int_\mathbb{R} \frac{1}{\pi(1 + (x-\theta)^2)} d\theta$$
Standard computation yields
$$m(x) = 1, \forall x \in \mathbb{R}$$
Even though it is not a valid density, notice that for any $\theta \in \Theta$, for *any* observed value x_0, $\mathbb{P}_\theta(m(X) \leq m(x_0)) = 1$ since the event $m(X) \leq m(x_0)$ is always trivially true with probability 1. Thus, if one checks for prior-data conflict in this problem, we will always get 1, implying that there is *no* prior-data conflict regardless of what we observed, indicating that generally, improper priors cannot be verified and checked via prior data conflict. □

In some sense, every component of our "model" for inference should be checkable. That is, our prior and model should be able to undergo diagnostics to check for its capability with data. The failure of improper priors to be examined using prior data conflict reflects a bigger problem in the general framework of improper priors from a philosophical point of view, that is, the "objectivity" of improper priors in scientific practice.

In Karl Popper's work *Conjectures and Refutations* (1963) [Pop63], he argued that "the criterion of the scientific status of a theory is its falsifiability, or refutability, or testability." In a statistical problem, under Popper's framework, one must be able to falsify the "ingredients" of inferences in order for it to be a valid scientific approach. If an improper prior could not be falsified through methods like checking for prior-data conflicts, its role in being an "objective" choice of prior in statistical inference would become questionable.

The objectivity of improper priors under the Popperian framework could also be challenged from an alternative point of view. Recall that we have demonstrated that ad-hoc modifications on Jeffreys priors and refererence priors are needed depending on different contexts. In the popperian framework, such ad-hoc modifications are reminiscent to a concept by Karl Popper, called *immunizing strategems* (See [Ost24]), meaning that they are being "modified" to be immune from refutation and being falsified, which in his view, will cause such improper prior theories to be labeled as pseudoscientific practices rather than scientific. While the judgment on whether it is indeed pseudo-science it depends on one's perspective in the philosophy of science, this undoubtedly further questions the scientific validity and the choice of using improper priors as an "objective" choice in statistical inference.

6. Conclusion and Further Directions

Throughout this paper, we have investigated several different perspectives to answer the question of whether or not improper priors are "objective". We can observe that, for simple priors such as uniform priors, it suffers from a lack of invariance under reparameterization and relabeling of parameters. To the other end, while improper priors such as reference priors are considered mathematically justified and optimal in an information-theoretic sense, it still depends on subjective choices, such as the ordering of parameters. In addition, due to challenges in implementation, one might resort to default priors such as the Jeffreys prior but as we have shown, this still depends on subjective choices regarding the "route" to obtain the Jeffreys prior on higher dimensional problems. In summary, combining these observations suggests that finding the "objective" improper prior is at heart, a constrained optimization problem, since it is fundamentally a tradeoff between "objectivity", practicality, and optimality. Unfortunately, this generally cannot be achieved unless we prioritize our choices to one of the criteria above. We have also demonstrated that improper priors suffer from an inability to undergo diagnostics such as prior-data conflict, which further challenges their role as "objective" tools in statistical inference.

Furthermore, we have also stylistically used quotation marks around the word objectivity whenever it appears in this paper. Personally, one of the biggest reasons why a truly "objective" improper prior has not yet been derived in statistical theory is because the word "objective" has no formalized definition in a unified sense. As an example, we have shown that "objectivity", epistemologically, depends on one's statistical interpretation of the word. To illustrate, one might think that information-theoretically optimal is an objective solution to the problem, yet another may prioritize decision-theoretic optimality. Philosophically, there is no universally agreed-upon definition of the word "objectivity" yet, which may be a crucial reason why a universally agreed answer to the problem has not been proposed since it is partly ill-posed.

While this problem may not admit a complete solution under current statistical theory, a potential, partial solution to the problem may be achieved via *data splitting*. To the best of our knowledge, while there are currently no extensive literature on this topic [4], the approach of data splitting is in spirit similar to cross-validation methods, and can be summarized into two broad steps:

(1) We first partition our data into two parts.
(2) Then, we use the former to construct a prior, and the latter for inference

This approach enables us to perform checks on priors via prior-data conflict, which was one of the largest barrier previously. However, the proportions in our data splitting still remains a subjective choice, and thus may only provide a partial solution to the problem.

[4] As discussed during one of the office hours.

Appendix

Example 5.3. We will now calculate $f(\overline{x_n})$ under H_a. That is, we will calculate

$$f(\overline{x_n}) = \int_{\mathbb{R}} f_\mu(\overline{x_n})\pi_{\text{Jeff}}(\mu)d\mu = \int_{-\infty}^{\infty} f_\mu(\overline{x_n})d\mu$$

since $\pi_{\text{Jeff}}(\mu) \propto 1$ so without loss of generality we will use $\pi_{\text{Jeff}}(\mu) = 1, \forall \mu \in \mathbb{R}$. Writing out the density, we get

$$\int_{-\infty}^{\infty} f_\mu(\overline{x_n})d\mu = \int_{-\infty}^{\infty} \frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{n}{2}\left(\overline{x_n} - \mu\right)^2\right) d\mu \tag{6.1}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\frac{1}{n}}} \exp\left(-\frac{(-1)^2(\mu - \overline{x_n})^2}{\frac{2}{n}}\right) d\mu \tag{6.2}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\frac{1}{n}}} \exp\left(-\frac{1}{2} \cdot \frac{(\mu - \overline{x_n})^2}{\frac{1}{n}}\right) d\mu \tag{6.3}$$

But notice that the integrand in Equation 6.3 is the density of $\mathcal{N}\left(\overline{x_n}, \frac{1}{n}\right)$. Thus, by the definition of a density, it is immediate that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\frac{1}{n}}} \exp\left(-\frac{1}{2} \cdot \frac{(\mu - \overline{x_n})^2}{\frac{1}{n}}\right) d\mu = 1$$

Hence,

$$f(\overline{x_n}) = 1 \text{ under } H_a. \tag{6.4}$$

$\square$

References

[BB91] James O. Berger and José M. Bernardo. On the development of the reference prior method. Technical Report Technical Report No. 91-15C, Department of Statistics, Purdue University, April 1991.

[BB92] James O. Berger and José M. Bernardo. Ordered group reference priors with application to the multinomial problem. *Biometrika*, 79(1):25–37, 1992.

[BBS09a] James O. Berger, José M. Bernardo, and Dongchu Sun. Objective priors for discrete parameter spaces. *Annals of Statistics*, 37(2):905–938, 2009. Available at: https://www2.stat.duke.edu/ berger/papers/discrete.pdf.

[BBS09b] James O. Berger, José M. Bernardo, and Dongchu Sun. The formal definition of reference priors. *The Annals of Statistics*, 37(2):905–938, 2009.

[BS94] José M. Bernardo and Adrian F. M. Smith. *Bayesian Theory*. Wiley Series in Probability and Statistics. John Wiley & Sons, Chichester, 1994.

[CB94] Bertrand S. Clarke and Andrew R. Barron. Jeffreys' prior is asymptotically least favorable under entropy risk. *Journal of Statistical Planning and Inference*, 41(1):37–60, 1994.

[EG21] Michael Evans and Yuling Guo. Measuring and controlling bias for some bayesian inferences and the relation to frequentist criteria. *Entropy*, 23(2):190, 2021.

[EM06] Michael Evans and Hila Moshonov. Checking for prior-data conflict. *Bayesian Analysis*, 1(4):893–914, 2006.

[Eva16] Michael Evans. Measuring statistical evidence using relative belief. *Computational and Structural Biotechnology Journal*, 14:91–97, 2016.

[Eva26] Michael Evans. Theory of statistical inference: Lecture vi.3, 2026. STA422/2162, University of Toronto.

[Jef86] Harold Jeffreys. *Theory of Probability*. Oxford University Press, 1986.

[MR99] Rahul Mukerjee and Nancy Reid. On a property of probability matching priors: Matching the alternative coverage probabilities. *Biometrika*, 86(2):333–340, 1999.

[Ost24] Aleksander Ostapiuk. Neoclassical economics' immunisation strategies against behavioural economics: Popper's perspective. *Gospodarka Narodowa / The Polish Journal of Economics*, 4(320):51–73, 2024.

[Pop63] Karl Popper. *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge, 1963.

[WP63] B. L. Welch and H. W. Peers. On formulae for confidence points based on integrals of weighted likelihoods. *Journal of the Royal Statistical Society. Series B (Methodological)*, 25(2):318–329, 1963.